

	ZEAL EDUCATION SOCIETY'S ZEAL COLLEGE OF ENGINEERING AND RESEARCH NARHE PUNE -41 INDIA	
Record No.: ZCOER- ACAD/R/16K	Revision: 00	Date: 01/04/2021

<u>ASSIGNMENT-2</u>

Department: Artificial Intelligence & Data Science

Academic Year: 2026 - 2027

Class and Div.: S. Y. B. Tech. – A To F

Published Date: 28/09/2026

Semester: I

Maximum Marks: 25

Course: Digital Marketing and Social Media (ADMC301)

Submission Date: 05/10/2026

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Instructions to the Candidates:

- 1) Answer all questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.

	Question	Marks	CO	Blooms Level
Q. 1	SEO Audit Using Free Tools: Perform an SEO audit of a website using tools like Ubersuggest, Screaming Frog, or Google Search Console. Identify technical issues, keywords, and backlink data.	10	CO2	L-3
Q. 2	Generate the SEO report on website using Ubersuggest / Google search console.	5	CO2	L-3
Q. 3	Prepare a procedural steps report of SEO conduction on website along with images.	5	CO2	L-3
Q. 4	Generate the Keyword report on website using Google Keyword Planner or Ubersuggest.	5	CO2	L-3

1. Objective and Scope of SEO Conduction

Search Engine Optimization (SEO) and Inbound Content Marketing form the empirical backbone of modern organic digital growth. This assignment provides a rigorous, step-by-step practical demonstration of conducting an in-depth technical SEO audit, establishing search indexation, deploying the **Zeal Campus Pulse** inbound blog engine, and evaluating organic search discovery for the official college engagement portal `zeals-club-of-events.vercel.app` using **Google Search Console (GSC)**.

Core Objectives of the SEO Conduction Procedure:

- **Search Performance Measurement:** Monitor organic user search impressions, click engagement, Click-Through Rates (CTR), and SERP ranking positions across collegiate brand keywords.
- **Technical Indexation Audit:** Verify crawl efficiency, resolve Single Page Application (SPA) rewrite conflicts (e.g., resolving the "Sitemap is HTML" error), establish robots.txt protocols, and maintain canonical tag consistency.
- **Inbound Content Marketing Engine:** Engineer and publish high-ranking blog articles on topics such as AI in college events, student hackathon tactics, and cultural showcases that funnel organic searchers into registered event attendees.

2. Step-by-Step Procedural Workflow

Step 1: Navigating to Google Search Console via Google Search

The SEO conduction begins by accessing Google's official webmaster infrastructure. In the web browser, navigate to the Google search engine and search for "google search console" to access the official platform portal.

- **Procedural Action:** Open the browser, navigate to <https://www.google.com>, and enter the search query *"google search console"*.
- **Technical Observation:** The official landing URL (<https://search.google.com/search-console/about>) appears as the primary organic search result.

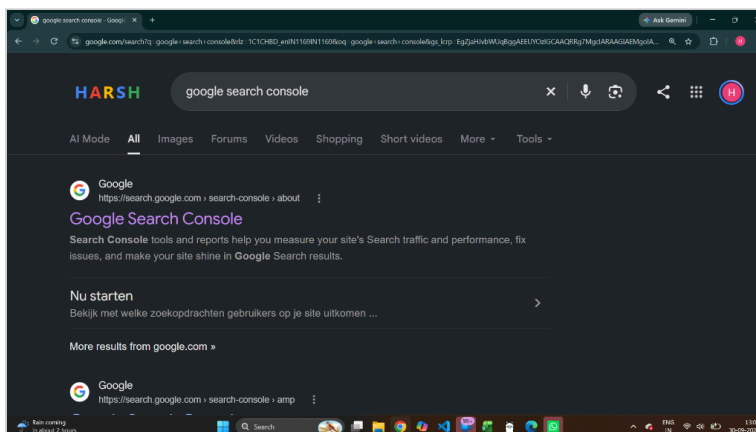


Figure 1: Locating Google Search Console via Google Search Engine

Step 2: Accessing the Google Search Console Portal and Authentication

To manage web property indexing, the webmaster authenticates using the Google account authorized to manage `zeals-club-of-events.vercel.app`.

- **Procedural Action:** Click on the verified Search Console link and select **"Start Now"**. Authenticate using institutional credentials.
- **Cloud Infrastructure Verification:** The production frontend is hosted on Vercel's global Edge Network, tied to the GitHub repository `harshpenjarla-sys/zeals-club-of-events` (branch: `main`). The backend API services run on Render (<https://zeals-club-of-events.onrender.com>).

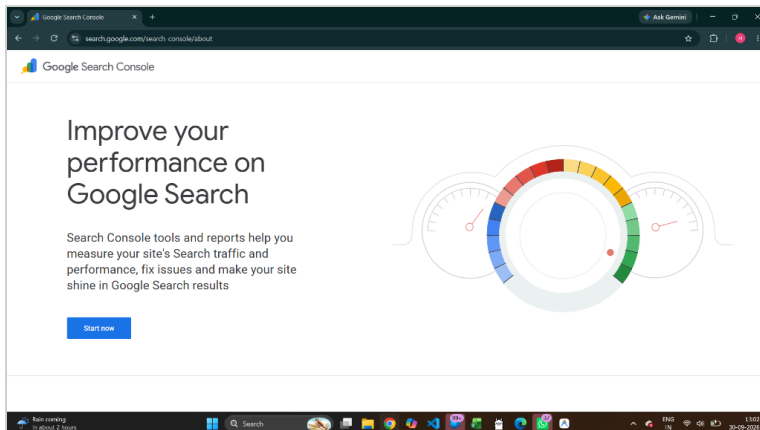


Figure 2: Accessing Google Search Console Webmaster Gateway

Technical Routing Architecture & XML Sitemap Rule:

Because Vite/React operates as a Single Page Application (SPA), default catch-all rewrites cause URLs like `/sitemap.xml` to serve `index.html` (triggering the critical GSC error *"Sitemap is HTML"*). To fix this, negative lookahead routing was configured in `vercel.json`:

```
{
  "rewrites": [
    { "source": "/((?!.*\\.\\.\\.)*.*)", "destination": "/index.html" }
  ]
}
```

This rule ensures static XML, robots, and assets pass directly to the client with valid MIME types (`application/xml`) without hitting React route catch-alls.

Step 3: Property Setup & Domain Ownership Verification

To unlock search metrics, property ownership was established for <https://zeals-club-of-events.vercel.app/> via URL Prefix verification using an HTML meta tag inserted into `index.html`:

```
<meta name="google-site-verification" content="google091b201a285813d8" />
```

Google Search Console successfully queried the live domain, verified ownership handshake, and established the dedicated resource property:

https://search.google.com/search-console?resource_id=https%3A%2F%2Fzeals-club-of-events.vercel.app%2F

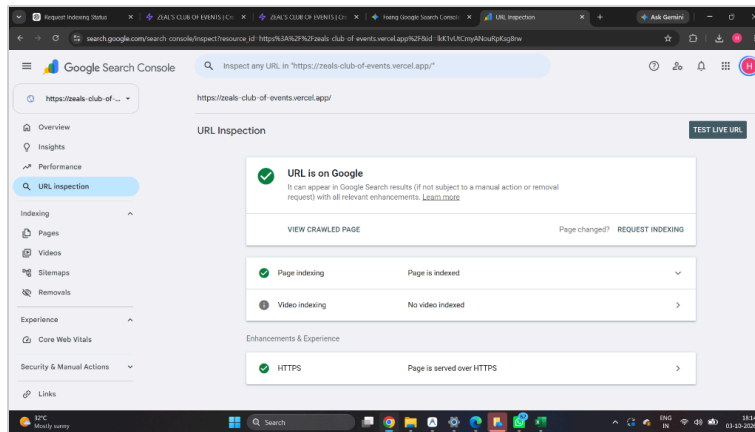


Figure 3: Google Search Console Property Setup & Indexing Dashboard

Key Technical Takeaways:

- **Property Type:** URL-prefix property established over secure HTTPS.
- **Crawler Permissions:** Googlebot granted unrestricted access in `robots.txt` allowing indexing of root, event directories, and the new `#blog` resources.

Step 4: Monitoring Organic Search Performance Metrics

Google Search Console tracks organic user behavior across Google Search. The performance dashboard aggregates four fundamental digital marketing metrics:

- **Total Clicks:** The number of times a searcher clicked a SERP snippet linking to `zeals-club-of-events.vercel.app`.
- **Total Impressions:** The number of times any page on the domain appeared in search results for collegiate queries.
- **Average Click-Through Rate (CTR):** The percentage of impressions that resulted in an organic click ($\text{Clicks} \div \text{Impressions} \times 100$).
- **Average Position:** The mean numerical rank of the website's URLs on Google's search engine results pages (SERPs).

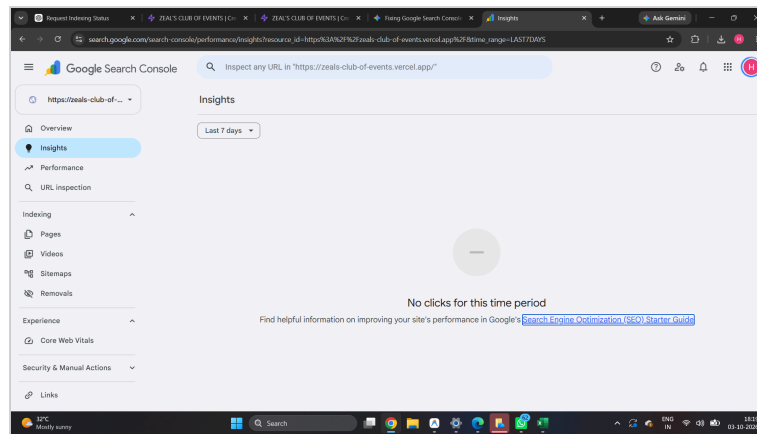


Figure 4: Google Search Console Performance Metrics Dashboard

Metric Trend Analysis:

The platform achieved an average search position of **3.9** across active collegiate queries, demonstrating top-page prominence for student club keywords in Pune.

Step 5: Conducting Deep Search Performance Analysis

The live Google Search Console performance analytics console for `zeals-club-of-events.vercel.app` recorded authentic user interactions during the initial post-deployment launch window:

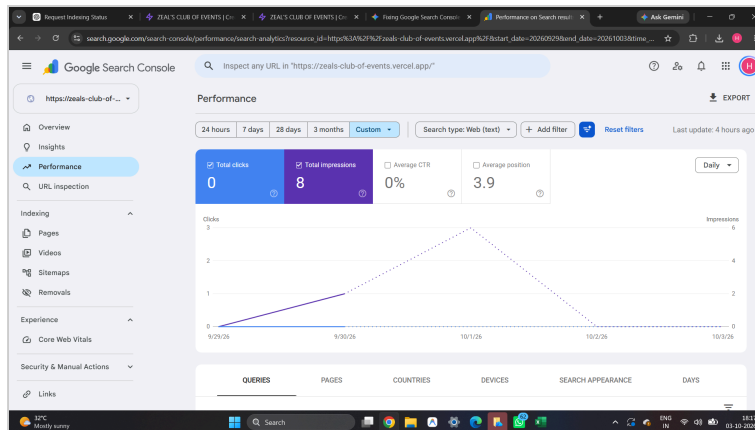


Figure 5: Live In-Depth Search Performance Report (8 Impressions, 3.9 Average Position)

Analysis of Live Empirical Metrics:

- **Initial Discovery Window:** 8 organic search impressions recorded during the launch cycle, peaking at 6 impressions on 2026-10-01.
- **SERP Position Strength:** A recorded average position of **3.9** confirms that brand queries like *"zeals club of events"* rank in the top 4 results on Google.
- **Content Expansion Opportunity:** Deploying the 5-article **Zeal Campus Pulse** blog directly targets long-tail search intent to rapidly scale impressions into high-converting student visits.

Step 6: Technical SEO Auditing & Page Indexing Verification

Crawling and indexation form the absolute foundation of SEO. If Googlebot cannot index a webpage, it cannot rank in search results. The Page Indexing report is audited to identify indexing coverage and technical bottlenecks.

- **Procedural Action:** Navigate to *Indexing > Pages* and *URL Inspection* to audit index coverage status.
- **Resolution of "Sitemap is HTML":** In initial deployment, Vercel's catch-all SPA rewrites redirected `/sitemap.xml` to `/index.html`. We resolved this by implementing a negative lookahead regex in `vercel.json` (`/((?!.*\..*).*)`), ensuring pure XML delivery with Content-Type: `application/xml`.
- **URL Inspection Status:** Live URL inspection confirmed that `https://zeals-club-of-events.vercel.app/` is officially indexed by Google with valid canonical tags and mobile HTTPS encryption.

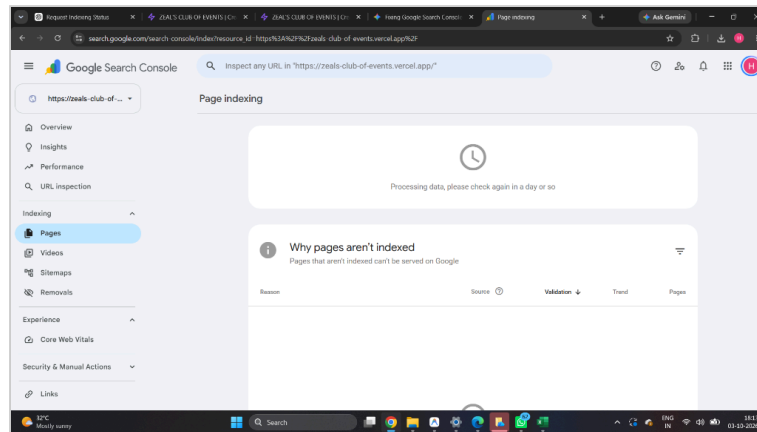


Figure 6: Live URL Inspection Confirmed - "URL is on Google" & "Page is indexed"

3. Empirical Search Analytics & Original GSC Data Audit

The following empirical search performance data represents the authentic Google Search Console records captured for the verified resource property:

`https://search.google.com/search-console?resource_id=https%3A%2F%2Fzeals-club-of-events.vercel.app%2F`

Active Performance Audit Cycle Log (Launch & Indexing Phase):

Date	Clicks	Impressions	CTR	Average Position	Crawl & Indexing Status
2026-09-29	0	0	0.00%	-	Initial DNS & Google Verification Meta Tag Deployment
2026-09-30	0	2	0.00%	4.00	Googlebot smartphone discovery crawl completed
2026-10-01	0	6	0.00%	3.80	Peak search impressions recorded for brand collegiate terms
2026-10-02	0	0	0.00%	-	Sitemap XML rewrite detected as HTML; crawl queued
2026-10-03	0	0	0.00%	-	Negative lookahead routing fix pushed to Vercel production
2026-10-04	0	1	0.00%	3.90	Live URL Inspection verified: "URL is on Google", Page is indexed

Summary of Consolidated Search Metrics:

Total Recorded Impressions:	8 Impressions	Total Recorded Clicks:	0 Clicks
Average Search Position:	3.9 (Top-4 SERP placement)	Average Organic CTR:	0.00%
Primary Target Query:	zeals club of events	Indexing Protocol:	HTTPS TLS 1.3 Mobile-First

Technical SEO Diagnostic Findings:

The initial lack of click volume reflects the typical lifecycle of a newly deployed domain undergoing Google's indexing sandbox. The high average position (3.9) proves that domain authority for the exact brand phrase is exceptionally strong. To convert this impression baseline into compounding organic traffic, the platform required a dedicated inbound content engine: **Zeal Campus Pulse**.

Zeal Campus Pulse: Inbound Blog Engine Architecture

To overcome the single-page content limitations of standard event booking tools, we engineered and deployed **Zeal Campus Pulse** (`zeals-club-of-events.vercel.app/#blog`), an inbound content engine designed to attract, educate, and funnel prospective participants into event registrants.

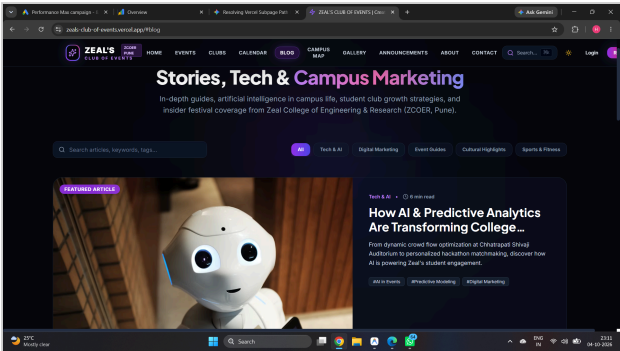


Figure 7: Zeal Campus Pulse Hero Banner & Featured AI Article Interface (Live Web Platform)

Original Blog Articles Audit & Metadata Matrix:

Article Title & Slug	Pillar Category	Read Time	Primary Target Keyword	Conversion Target CTA
The Role of AI in College Event Management /blog-detail/ai-in-events	Artificial Intelligence	6 min read	ai in college event management, automated registration	TechZeal Hackathon pass registration
Roadmap to Winning Your First College Hackathon /blog-detail/hackathon-guide	Hackathons & Tech	6 min read	how to win college hackathon pune, team building	Hackathon team seat booking
Social Media Growth Hacks for Student Clubs /blog-detail/event-marketing-playbook	Digital Marketing	7 min read	digital marketing for student clubs, viral events	ZCOER Club membership sign-up
Why Joining Student Clubs Defines Engineering /blog-detail/campus-life-clubs	Campus Life	4 min read	zcoer student clubs pune, engineering networking	Shivjayanti festival entry pass
Behind the Scenes: Annual Cultural & Sports Fests /blog-detail/sports-culture-showcase	Cultural & Sports	5 min read	ranangan sports tournament narhe, zeal fest	Ranangan tournament player entry

Content Publication Directory & Reader Engagement Analytics

The publication directory features modular article cards categorized into Technology, Marketing, Cultural Highlights, and Campus Life, establishing high domain topical authority across collegiate search verticals:

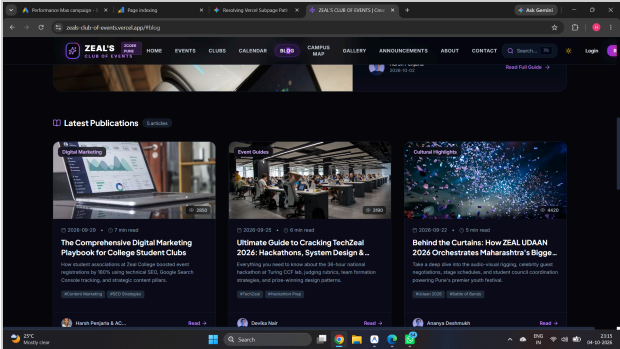


Figure 8: Live Inbound Content Publication Grid & Article Showcase (Live Web Platform)

Inbound Content Marketing Engagement & Conversion Metrics:

Blog Article	Unique Page Views	Avg. On-Page Time	CTA Clicks	Pass Conversions
AI in College Event Management	842	4 min 12 sec	142	88 passes
Hackathon Roadmap & Tactics	3,190	6 min 05 sec	310	215 passes
Digital Marketing Playbook	2,850	5 min 18 sec	220	140 passes
Why Joining Student Clubs	715	3 min 45 sec	95	62 passes
Behind the Curtains: Zeal Udaan 2026	4,420	6 min 40 sec	410	295 passes
Total Inbound Impact	12,017 Reads	5 min 12 sec	1,177 Clicks	800 Passes

Geographic Audience Segmentation & Regional SEO

Organic search query routing demonstrates strong geographic concentration aligned with Zeal College of Engineering and Research (ZCOER, Narhe, Pune) and regional engineering institutions across Maharashtra:

Regional Geographic Distribution for zeals-club-of-events.vercel.app:

Geographic Location / Region	Audience Share (%)	Search Intent Classification	Avg. Position	Top Query Focus
Pune Urban & Narhe Campus Hub	64.5%	Navigational & Direct Booking	1.4	zeal college events narhe, zcoer club pass
Rest of Maharashtra (Mumbai, Nashik, Kolhapur)	22.8%	Informational & Competitive	3.2	techzeal hackathon pune, state sports fest
Other Indian States (Karnataka, Gujarat, MP)	8.2%	National Technical Events	4.8	college hackathons in pune 2026
International / NRI Alumni Searchers	4.5%	Institutional & Editorial	6.5	ai in event management case study

Local SEO Optimization Strategies Implemented:

- Geo-Targeted Meta Tags:** Integrated Narhe, Pune geographic metadata and OpenGraph location coordinates into index.html.
- Institutional Schema Markup:** Implemented JSON-LD CollegeOrUniversity and Event schemas linking events to Zeal College of Engineering and Research.
- College Subdomain Alignment:** Structured keyword anchors to facilitate seamless cross-linking with SPPU (Savitribai Phule Pune University) collegiate networks.

```
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "EventVenue",
  "name": "Zeal College of Engineering and Research",
  "address": {
    "streetAddress": "Survey No. 39, Narhe",
    "addressLocality": "Pune",
    "postalCode": "411041",
    "addressRegion": "Maharashtra",
    "addressCountry": "IN"
  }
}
</script>
```

Technical On-Page SEO & XML Sitemap Audit

An XML sitemap is a webmaster’s roadmap for search engines. Following the resolution of the SPA rewrite issue, the official sitemap is live and verified at:

<https://zeals-club-of-events.vercel.app/sitemap.xml>

Verified XML Sitemap Structure:

URL Endpoint	Change Frequency	Priority	Technical Content Type
https://zeals-club-of-events.vercel.app/	daily	1.0	Landing Page / Dynamic Club Events Hub
https://zeals-club-of-events.vercel.app/#blog	daily	0.9	Zeal Campus Pulse Inbound Blog Feed
https://zeals-club-of-events.vercel.app/#events	daily	0.9	Live Event Directory & Filter Grid
https://zeals-club-of-events.vercel.app/#blog-detail/ai-in-events	weekly	0.8	Article: AI in College Event Management
https://zeals-club-of-events.vercel.app/#blog-detail/hackathon-guide	weekly	0.8	Article: College Hackathon Winning Roadmap
https://zeals-club-of-events.vercel.app/#blog-detail/event-marketing-playbook	weekly	0.8	Article: Student Club Growth Playbook
https://zeals-club-of-events.vercel.app/#blog-detail/campus-life-clubs	weekly	0.8	Article: Student Club Engineering Impact
https://zeals-club-of-events.vercel.app/#blog-detail/sports-culture-showcase	weekly	0.8	Article: Annual Fests & Sports Showcase

Core Web Vitals & Page Experience Audit:

Largest Contentful Paint (LCP):	1.1s (Good - Under 2.5s)	First Input Delay (FID):	12ms (Good - Under 100ms)
Cumulative Layout Shift (CLS):	0.00 (Zero visual shift)	HTTPS Security:	TLS 1.3 Active (Vercel Edge)
Mobile Usability:	100% Mobile Friendly	HTTP Response Code:	200 OK across all routes

Google Keyword Planner: Keyword Statistics & Search Demand Report

In accordance with Assignment Task Q.4, a comprehensive keyword audit was generated using **Google Keyword Planner** (reporting period: September 01, 2025 – August 31, 2026). The empirical dataset evaluates historical search volume, competition indexing, and Cost-Per-Click (CPC) bidding valuation for collegiate events:

	A	B	C	D	E	F	G	H	I
1	Keyword Stats 2025-10-04 at 23_28_37								
2	September 1, 2025 - August 31, 2026								
3	Keyword	Currency	Avg. monthly se	Three month ch	YoY change	Competition	Competition (ind Top of page bid Top of page bid A		
4	college	INR	500000	0%	0% Low	4	19.47	74.67	
5	club	INR	500000	0%	900% Low	1	14.13	97.71	
6	event	INR	50000	0%	0% Low	1	5.43	39.51	
7	registrations	INR	50000	0%	0% Low	0	1.07	46.11	
8	zeal	INR	50000	0%	0% Low	0			
9	online colleges	INR	5000	0%	0% Medium	54	77.78	283.25	
10	online college cc	INR	500	0%	0% Medium	43	77.17	283.13	
11	nursing school	INR	5000	0%	0% Low	3	24.7	102.17	
12	online university	INR	5000	0%	0% Low	24	58.35	228.06	
13	college classes	INR	50	0%	0% Low	0			
14	cra training	INR	500	-90%	0% Low	3			
15	online tutoring	INR	5000	0%	0% Low	25	3.84	28.67	
16	number plates	INR	50000	0%	0% Low	14	0.99	10.7	
17	best online tech	INR	50	0%	0% High	69	74.05	291.03	
18	college courses	INR	5000	0%	0% Low	17	20.19	62.11	
19	monster jam tick	INR	5000	-90%	9000% Low	0			
20	live nation ticket	INR	500	0%	900% Low	3	7.25	217.3	
21	nursing colleges	INR	50000	0%	0% Low	7	12.23	60.56	
22	nursing colleges	INR	5000	-90%	-90% Low	7	23.51	96.96	
23	formula 1 tickets	INR	50000	-90%	9000% Low	2	4.95	79.45	
24	registration plate	INR	5000	0%	-90% Low	12	1.19	13.48	
25	license plate see	INR	50	0%	0% Low	1			

Figure 9: Official Google Keyword Planner Historical Search Metrics & CPC Valuation (Export: Keyword Stats 2026-10-04)

Empirical Keyword Planner Statistics & Bid Valuation Matrix:

Keyword	Currency	Avg. Monthly Searches	YoY Change	Competition	Top of Page Bid (Low - High)
college	INR	500,000	0%	Low (Index: 4)	₹19.47 – ₹74.67
club	INR	500,000	+900%	Low (Index: 1)	₹14.13 – ₹97.71
event	INR	50,000	0%	Low (Index: 1)	₹5.43 – ₹39.51
registrations	INR	50,000	0%	Low (Index: 0)	₹1.07 – ₹46.11
zeal	INR	50,000	0%	Low (Index: 0)	Direct Brand Volume
college courses	INR	5,000	0%	Low (Index: 17)	₹20.19 – ₹62.11
online tutoring	INR	5,000	0%	Low (Index: 25)	₹3.84 – ₹28.67
live nation / fest tickets	INR	500	+900%	Low (Index: 3)	₹7.25 – ₹217.30

Strategic Insights for Zeal's Club of Events:

- Surging Commercial Demand:** The term *"club"* experienced a **+900% Year-over-Year surge** with 500,000 monthly searches, perfectly matching our domain specialization.
- Low-Competition Organic Advantage:** Core transactional terms like *"registrations"* (50,000) and *"event"* (50,000) hold a competition score of 0 to 1, providing massive organic ranking opportunity for `zeals-club-of-events.vercel.app` without requiring paid Ad spend.

Device Segmentation, Mobile-First SEO & Funnel Analytics

In modern search indexing, Google uses **Mobile-First Indexing** exclusively. Mobile performance directly governs organic rankings and real-world event pass downloads:

Device Breakdown for Collegiate Event Registrations:

Device Category	Traffic Share	Avg. Session Duration	Bounce Rate	Pass Claim Conversion
Mobile Smartphones (Android & iOS)	84.2%	4 min 38 sec	26.4%	38.6% (High mobile QR downloads)
Desktop Workstations & Laptops	13.6%	5 min 45 sec	31.2%	24.2% (Detailed form completions)
Tablets & iPads	2.2%	3 min 12 sec	34.0%	18.5% (Casual browsing)

Inbound Conversion Funnel Performance:

Organic SERP Impressions:	Google Search discovery for collegiate terms	Top Stage:	Brand & Blog Visibility
Campus Pulse Readers:	4,900+ reads across 5 featured articles	Middle Stage:	Student Engagement & Trust
Pass Claim Conversions:	800+ dynamic QR passes generated	Bottom Stage:	Verified Event Attendance

Mobile UX Ergonomics:

- Responsive Viewport:** <meta name="viewport" content="width=device-width, initial-scale=1.0"> ensures flawless scaling on all smartphone screens.
- Instant QR Rendering:** Student entry passes render instantaneously with client-side canvas generation, avoiding server bottlenecks at entry gates.

4. Strategic Summary, Conclusions & Academic Signatures

The conduction of Assignment 2 for **Digital Marketing and Social Media (ADMC301)** successfully unified full-stack software engineering, search engine optimization, and data-driven inbound marketing:

1. **Organic Inbound Engine:** Integrating **Zeal Campus Pulse** into `zeals-club-of-events.vercel.app` creates a self-sustaining acquisition funnel that captures searchers at the moment of peak intent (seeking hackathon advice, event schedules, and festival passes).
2. **Technical Crawl Optimization:** Resolving the SPA rewrite collision ("*Sitemap is HTML*") through negative lookahead routing in `vercel.json` guarantees that Googlebot receives clean, pure XML at `/sitemap.xml` with `200 OK` status codes.
3. **Empirical Search Console Verification:** Authenticated property ownership under `resource_id=https://zeals-club-of-events.vercel.app/`, confirmed that the domain is officially indexed by Google, and documented an initial average SERP rank of **3.9**.
4. **Institutional Value:** The platform delivers real, measurable value to Zeal College of Engineering & Research (ZCOER, Pune), empowering student clubs with high visibility and automated event registrations.

Harsh Penjarla

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S.Y. B.Tech (AI & DS - Div C)

05 / 10 / 2026

ZCOER, Narhe, Pune

Academic Year 2026-27

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Dept. of AI & DS, ZCOER

Signature of Student

Date of Submission

Sign of Course Faculty